Diane-Marie Brache-Smith

Curriculum Vitae

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Department of Molecular and Cell Biology, University of California Merced

Education

2021–Present	**Ph.D., Quantitative and Systems Biology** (Expected August 2026) University of California Merced, Merced, CA Dissertation: *Integrating Cultivation, Multi-Omics, and Metabolic Modeling to Characterize the Zostera marina Microbiome for Seagrass Restoration* Advisor: Dr. Maggie Sogin GPA: 4.0
2023	**M.S., Quantitative and Systems Biology** University of California Merced, Merced, CA
2021	**B.S., Biological Sciences**, Biotechnology Concentration Miami Dade College, Miami, FL | Graduated with Honors
2019	**A.A., Biological Sciences**, Biotechnology Concentration Miami Dade College, Miami, FL

Publications

Peer-Reviewed

1. Badillo, J., **Brache-Smith, D.M.**, Sogin, E.M. (2025). Six complete genome sequences of a putative nitrogen-fixing bacteria from the genus *Agarivorans* sp. isolated from the seagrass, *Zostera marina*. *Microbiology Resource Announcements*. https://doi.org/10.1128/mra.00577-25

Submitted

2. **Brache-Smith, D.M.**, Badillo, J., Maeda, S., Sogin, E.M Putative Identification of bacterial candidates that promote the growth of the seagrass *Zostera marina Microbiome*. Submitted. Preprint: https://doi.org/10.64898/2026.03.19.712741

In Preparation

3. Brache-Smith, D.M., Chaput, G.M., Rojas, V., Badillo, Zabinski, K., J., Maeda, S., Sogin, E.M Integrating Metagenomics, Metatranscriptomics, and Metabolomics to Characterize Seasonal Nutrient and Phytohormone Dynamics in the *Zostera marina* Rhizosphere.

Awards, Fellowships & Grants

2023–2026	NSF Graduate Research Fellowship (GRFP), Award #21-39297 ($161,000)
2026	Molecular and Cell Biology Award for Outstanding Graduate Student, UC Merced ($75)
2026	Best Student Poster, EMSL–JGI User Meeting ($200)
2025	JGI Genomic Encyclopedia of Bacteria and Archaea (GEBA IV) Award – PacBio Sequencing ($9,000)

2025	Gordon Research Conference Applied Environmental Microbiology Poster Prize & Early Career Scientist Award ($500)
2025	QSB Travel Fellowship ($3,900)
2024	MacKenzie Scott Graduate Student Travel Award ($1,000)
2024	QSB Travel Fellowship ($750)
2023	Marine Biological Laboratory Alumni ROCs (Regional Outreach and Communication in STEM) Award ($2,500)
2023	Ford Foundation Fellowship Alternate
2022	Ford Foundation Fellowship Honorable Mention
2022	Department of Energy Microbial Diversity Course Award ($9,500)
2022	QSB Graduate Research Improvement Award ($1,000)
2021	NSF/UF Pathways to Success Scholarship ($3,600)
2019–2021	Bank of America STEM Scholarship ($1,800, multiple awards)
2020	Great Minds in STEM Scholar ($1,000)
2019	USDA iCATCH Grant ($3,800)
2019–2021	Beta Beta Beta Honor Society, Vice President
2015	Miami Swim Week Peroni Fashion Design Contest Finalist
2010	Miami New Times Top 100 Creatives #67 — Miami's Cultural Superheroes
2010	Miami New Times Best of Miami — Best Designer of Retro Threads

Workshops

Summer 2022 — **Research Trainee**, Microbial Diversity Course

Marine Biological Laboratory, Woods Hole, MA

Designed and executed independent project creating sulfide-oxygen gradients to separate community niches

Created custom growth media to target bacteria, virus, protist and archaea of interest in anaerobic and aerobic conditions

Performed molecular, computational, and quantitative imaging analyses on protist, bacteria, and phage (single cell sorting, flow cytometry, FISH, TEM)

2024 — **JGI Microbial Genomics & Metagenomics Workshop**

Joint Genome Institute, Berkeley, CA

Presentations

Oral Presentations

Brache-Smith, D.M. (2026). Cultivating Solutions: Targeted Isolation, Identification, and Characterization of Zostera marina Associated Bacteria to Enhance Seagrass Resilience and Support Restoration Efforts. University of California Merced Molecular and Cell Biology Department Seminar, Merced, CA. **Pending Fall 2026*

Brache-Smith, D.M. (2024). La composición y el potencial funcional de bacterias promotoras del crecimiento dentro de la rizosfera de las fanerógamas marinas. *UC Merced Annual Symposium of Inclusive Excellence*, Merced, CA.

Brache-Smith, D.M. (2024). The composition and functional potential of growth promoting bacteria within the seagrass rhizosphere. *World Seagrass Conference & 15th International Seagrass Biology Workshop*, Napoli, Italy.

Brache-Smith, D.M. (2024). The composition and functional potential of growth promoting bacteria within the seagrass rhizosphere. *World Seagrass Conference & 15th International Seagrass Biology Workshop*, Napoli, Italy.

Brache-Smith, D.M. (2024). The Seagrass Microbiome from Marco to Micro, Bay Area RaMP Program in Microbiome Sciences, University of California Davis Bodega Bay Marine Lab, California

Brache-Smith, D.M. (2021). Basics of Social Media for Science, Miami Dade College Undergraduate Research Academy, Miami, Florida, United States

Brache-Smith, D.M., (2021) Applying Biological Anthropology and Chemistry to Ancient Roman Skeletons (MC and Moderator), Miami Dade College STEMversation with Dr. Kristina Killgrove, Miami, Florida, United States

Brache-Smith, D.M., Williams, A., Abdool, A., and Munoz, J. (2020). Examining Enzyme Replacement Therapy of CLN2 Disease Using a Human Neural Progenitor Cell Model, FIU McNair Research Conference, Miami, Florida

Brache-Smith, D.M., Williams, A., Abdool, A., and Munoz, J. (2020). Examining Enzyme Replacement Therapy of CLN2 Disease Using a Human Neural Progenitor Cell Model, Annual STEM Research Symposium 2020, Miami Dade College School of Science, Miami, Florida

Brache-Smith, D.M., Laas, P., Stingl, U., (2020). Culturing and Fluorescence in situ Hybridization of SAR 11 Bacteria, iCATCH Grant Web Presentation, Miami, Florida

Brache-Smith, D.M. (2020). Why Tutoring is a Great College Gig, Life Sciences South Florida Webinar, Virtual

Brache-Smith, D.M., Laas, P., Stingl, U., (2020). Culturing and Optimization of Fluorescence in situ Hybridization of SAR 11 Bacteria, Life Sciences South Florida STEM Undergrad Symposium, May 2020, FIU Biscayne Bay Campus, Miami, Florida, United States (Postponed due to COVID-19)

Brache-Smith, D.M., Martin, E., Griffith, K., (2015) Fuse Miami: Guest speaker on branding, social media marketing and how to make the most of current online selling platforms, MADE at The Citadel, November 2015, Miami, Florida

Brache-Smith, D.M., (2015) Frehling, J., Smelko, S., Making It: A forum for women in creative industries, Wolfsonian-FIU Museum, Miami Beach, Florida

Poster Presentations

Brache-Smith, D.M. (2026). Cultivating Solutions: Targeted Isolation, Identification, and Characterization of Zostera marina Associated Bacteria to Enhance Seagrass Resilience and Support Restoration Efforts. EMSL-JGI Joint User Meeting: Empowering Automated Laboratories—Integrating Experimentation with Data, Seattle, WA. **Winner Student Poster Prize*

Brache-Smith, D.M. (2025). Harnessing Beneficial Seagrass-Associated Bacteria: Targeted Cultivation, Genomic Exploration, and Accessing Plant Growth Promoting Abilities. *Gordon Research Seminar on Applied and Environmental Microbiology*, South Hadley, MA. **Winner Early Career Researcher Poster Prize*

Brache-Smith, D.M. (2024). Isolation of Seagrass Associated Bacteria with Plant Growth Promoting Traits. *JGI Microbial Genomics & Metagenomics Workshop*, Berkeley, CA.

Brache-Smith, D.M., Sogin, E.M. (2023). Exploring the Seagrass Microbiome for Potential Probiotic Growth-Promoting Sediment Amendments. *Joint Berkeley Initiative of Microbiome Sciences*, Berkeley, CA.

Brache-Smith, D.M. (2023). Capturing and quantifying beneficial bacteria of seagrasses. *UC Merced QSB Retreat*, Vista Ranch, CA.

Brache-Smith, D.M., Sogin, E.M., (2022) Exploring the Seagrass Microbiome for Potential Probiotic Growth-Promoting Sediment Amendments, Western Society of Naturalists, Oxnard, California

Research Experience

2021–Present **Graduate Research Fellow**, Sogin Lab

University of California Merced

Developed cultivation-based approaches to isolate plant growth-promoting bacteria from Zostera marina rhizosphere, rhizoplane, and endosphere (>400 isolates; 296 with PacBio/Nanopore whole genomes)

Integrated metagenomics, metatranscriptomics, and metabolomics with computational metabolic modeling to design minimal bacterial consortia for seagrass restoration

Designed and executed raceway inoculation experiments testing plant growth-promoting consortia on Zostera marina performance

2020–2021 **Undergraduate Researcher**, Laughinghouse Lab

University of Florida, Davie, FL

Characterized novel Scytonema species from Everglades periphyton mats using morphological and 16S-23S ITS rRNA analysis

2020 **Undergraduate Researcher**, Munoz Lab

Nova Southeastern University, Davie, FL

Examined enzyme replacement therapy effects on neural stem/progenitor cell proliferation and differentiation (CLN2 disease model) using TUNEL and EdU assay

Designed and implemented image analysis pipelines to quantify cell death and proliferation in human neural progenitor stem cell quantitative fluorescent microscopy

2019–2020 **Undergraduate Researcher**, Stingl Lab

University of Florida, Department of Microbiology & Cell Science

Cultivated SAR11 bacteria; optimized Fluorescence In Situ Hybridization staining protocols for bacteria *Polynucleobacter*

Research Skills

Wet Lab: Cultivation and isolation of cryptic and fastidious microbes, DNA/RNA extraction, PCR, quantitative microscopy (FISH, viral TEM), flow cytometry, phenotypic assays (nitrogen fixation, phosphate solubilization, sulfur oxidation, auxin production), micro electrodes (O_2, redox, H_2S) horizontal gene transfer and plasmid acquisition

Computational: Metagenomics, metatranscriptomics, genome assembly and annotation, metabolic modeling pipelines,16S rRNA analysis, R, Python, Snakemake, Pixi, Conda, Pathway Tools, High Performance Computing, Fiji, Cell profiler

Field: Marine water, benthic, sediment and rhizosphere sampling, wilderness first aid certified, field risk assessment certified, probiotic raceway experiments, PADI certified

Other: Curriculum development, science communication, bilingual (English/Spanish), graphic design (Adobe CS suite), autocad, sewing, carpentry, plumbing, electrical,

Mentoring

Undergraduate Researchers

2023–Present **Jackie Badillo**, UC Merced

URISE Fellow ($40,000 award); Summer research internship at UC Berkeley
Co-authored publication in *Microbiology Resource Announcements* (2025)

2023–Present **Saray Maeda**, UC Merced

URISE Fellow ($40,000 award); Summer research internship at Joint Genome Institute
Thermo Fisher Scientific Prize ($8,000)

2024 **RAMPS Microbiomes Project Guest Collaborator**

Multi-institutional research partnership with UC Berkeley (Corella Lab) and UC Davis Bodega Marine Laboratory

Teaching Experience

2023 **Teaching Assistant & Curriculum Developer**, Microbiome Research CURE

University of California Merced (Instructor of Record: Dr. Maggie Sogin)
Developed core laboratory curriculum: media design, DNA extraction, genome sequencing and assembly
Created bioinformatic analysis scripts

Led wet lab instruction and computational mentoring

2024 **Guest Lecturer**, Diverse Microbial Metabolisms CURE

Developed video lectures on microbial metabolisms and cultivation strategies

2022 **Teaching Assistant**, Introduction to Biology

University of California Merced

2021 **Teaching Assistant**, Nutrition

University of California Merced
Developed flipped classroom curriculum

2018–2021 **Chemistry Learning Assistant & STEM Lab Tutor**

Miami Dade College

Outreach and Professional Service

2026 **Workshop Facilitator** (Contracted), Fresno State University

Designed and delivered two 90-minute professional development workshops on undergraduate research barriers at Hispanic-Serving Institution (~180 participants total)
Led three-person facilitation team; funded by Center for Access to Science for All

2025 **Workshop Developer & Facilitator**, SACNAS National Conference

Created 75-minute workshop training faculty on inclusive mentorship and CURE design

2023 **Outreach Program Developer**, Microbe Hunters

Hayward High School (~40 students)

Designed hands-on program introducing microbial diversity through catalase assays and Foldscope microscopy; funded by Marine Biological Laboratory

2018–2020 **Citizen Scientist**

NOAA Phytoplankton Monitoring Network – HABs monitoring

Rescue a Reef, University of Miami – Coral nursery maintenance and transplantation

References

Upon Request